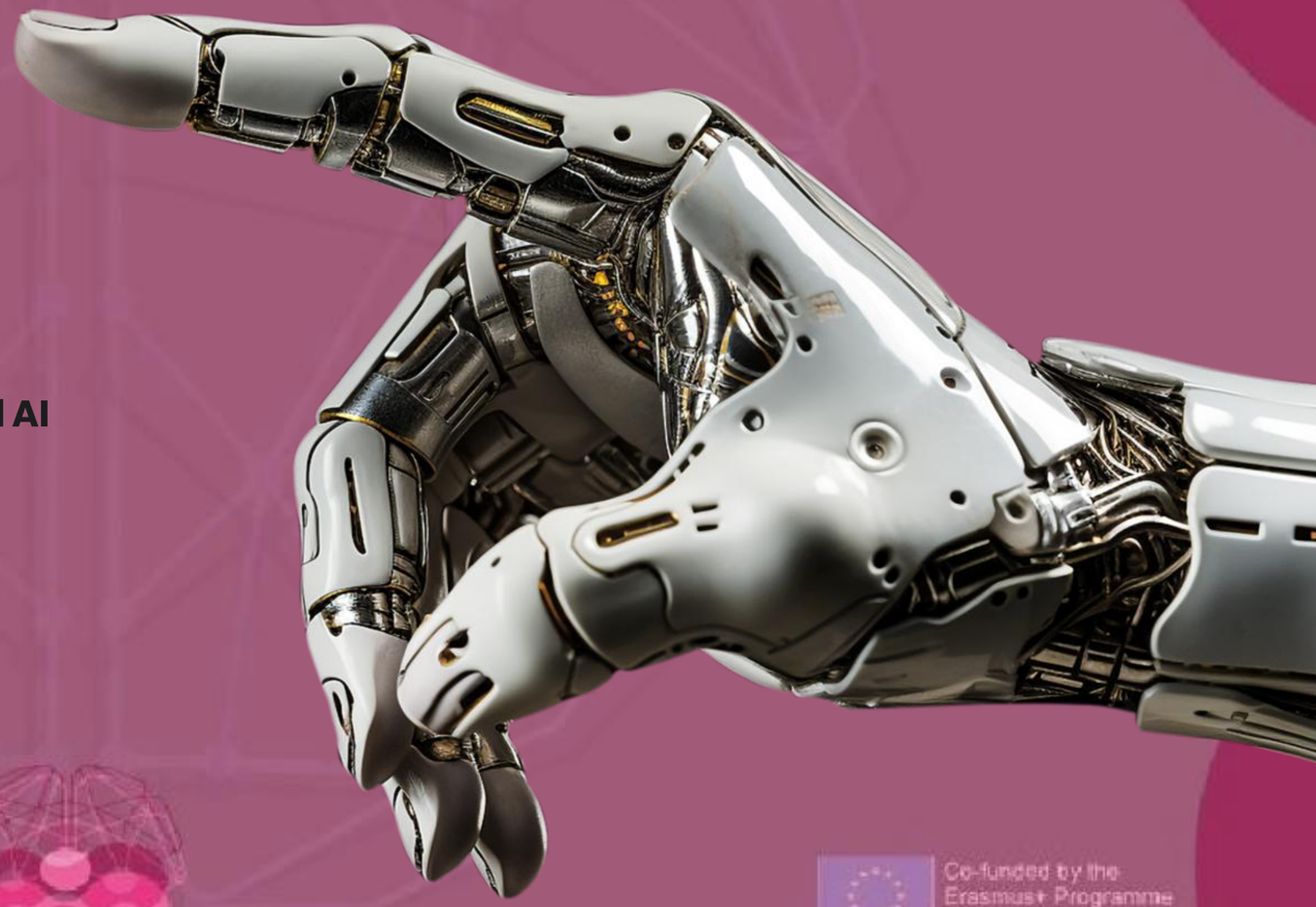
GENERATIVE AI

Design Thinking for Change



Transforming Hate Speech Challenges into Human-Centered AI Solutions Through a Structured, Empathetic Design Process

Module 2: AI Hackathon

Fighting Hate Speech Online | Innovation & Social Impact

Erasmus+

Co-funded by the Erasmus+ Programme of the European Union

Module Overview

01. Design Thinking Framework

Explore the 5-phase methodology: Empathize, Define, Ideate, Prototype, Test applied to hate speech detection challenges.

02. User-Centric Research

Develop deep empathy through personas, empathy mapping, and stakeholder analysis for comprehensive problem understanding.

03. Advanced AI Solutions

Discover cutting-edge ML frameworks, NLP techniques, and tool recommendations for building robust hate speech detection systems.

04. Rapid Prototyping & Iteration

Build, test, and refine AI-driven solutions using agile methods and data-driven insights for hackathon success.

The 5 Phases of Design Thinking

Phase 1: EMPATHIZE

Deeply understand users' pain points, needs, and contexts. Conduct interviews with content moderators, victims, and platform stakeholders to gather qualitative insights about hate speech impact.

Phase 2: DEFINE

Synthesize research into a clear problem statement. Develop How Might We (HMW) questions that reframe challenges into opportunities for AI-driven innovation.

Phase 3: IDEATE

Generate diverse solution concepts using brainstorming, design sprints, and collaborative workshops. Explore ML architectures, ensemble methods, and novel detection approaches.

Phase 4: PROTOTYPE

Rapidly build working AI models, MVPs, and interactive demos. Implement proof-of-concepts using pre-trained models, fine-tuned transformers, and ensemble techniques.

Phase 5: TEST

Validate solutions through user testing, A/B testing, and performance metrics. Iterate based on feedback, bias audits, and real-world deployment scenarios.

Phase 1: EMPATHIZE - Understanding Hate Speech

Deep Empathy Research

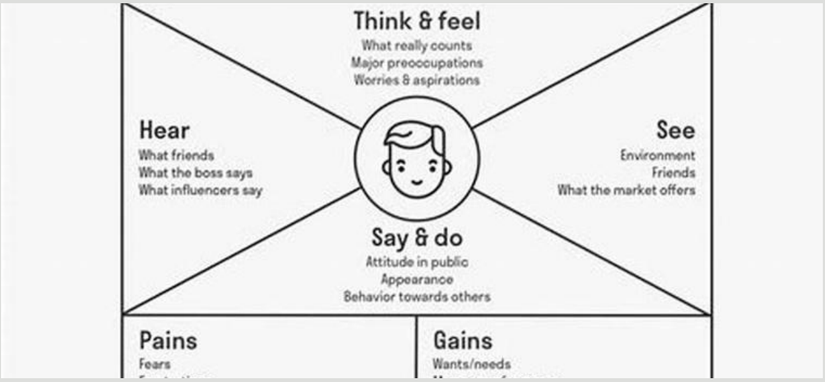
Move beyond data—understand human experiences. Interview content moderators facing psychological strain, marginalized communities experiencing harassment, and platform teams managing scale.

Stakeholder Mapping

Map all actors: victims, perpetrators, platforms, regulators, civil society. Understand competing needs and ethical considerations in content moderation.

Contextual Analysis

Study language nuances, cultural context, platform-specific dynamics. Hate speech varies by region, culture, and linguistic complexity—one-size-fits-all fails.



Phase 2: DEFINE - Problem Framing

Create User Personas

Develop detailed personas representing key stakeholder groups with specific needs, technical proficiency, and constraints.

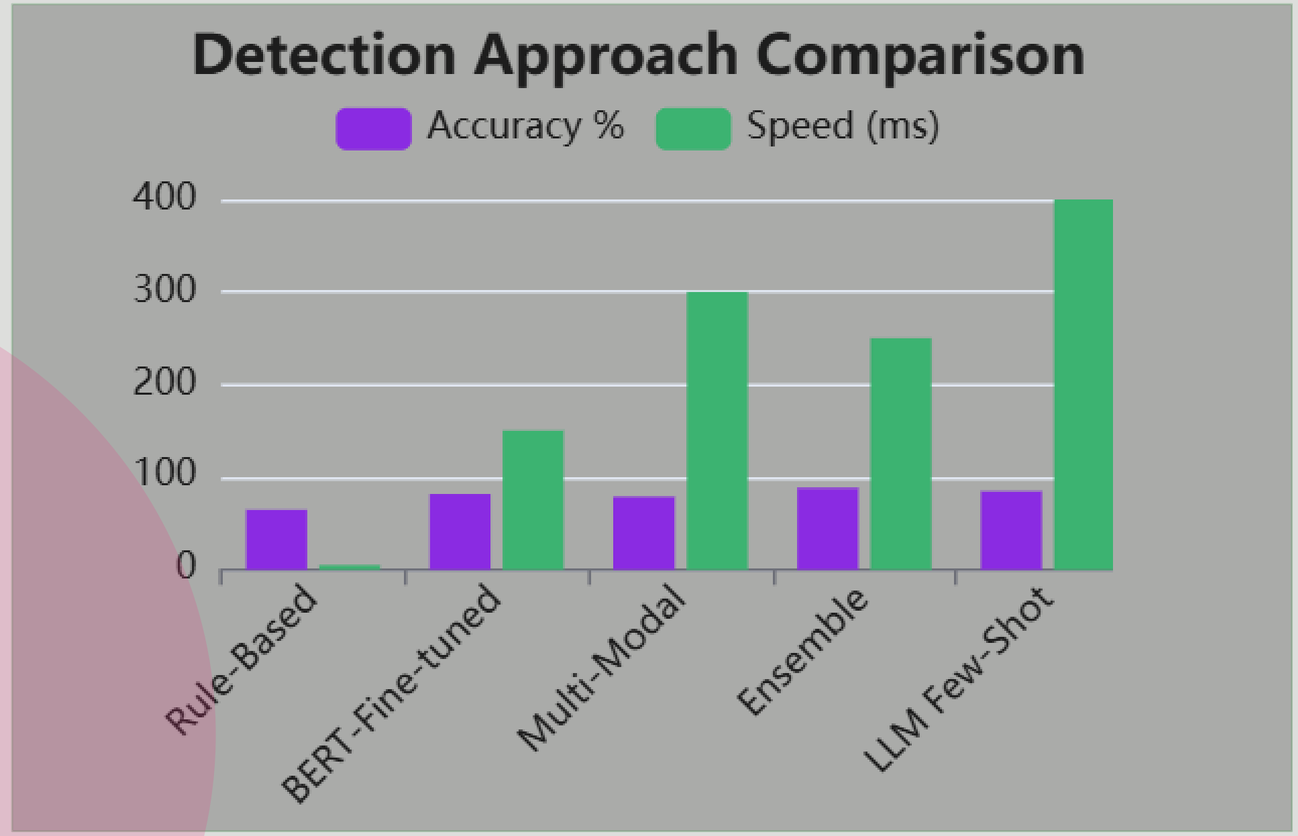
- Persona: Overburdened Moderator facing burnout from exposure to traumatic content.
- Persona: Marginalized User experiencing targeted harassment on multiple platforms simultaneously.
- Persona: Platform Product Manager balancing free speech with user safety at billion-user scale.

Develop HMW Questions

Transform insights into opportunity-focused problem statements that guide ideation and prototyping.

- HMW: Enable real-time detection that respects cultural context and linguistic nuance?
- HMW: Reduce human moderator burden while maintaining accuracy and reducing bias?
- HMW: Create transparent, explainable AI that users can understand and appeal?

Phase 3: IDEATE - AI Solution Concepts



Multi-Modal Detection

Combine text, image, video analysis. CNN for visual content, RNN for temporal patterns, transformers for semantic understanding.

Ensemble Methods

Stack multiple models (BERT, Gemma, DistilBERT, XLM-RoBERTa) for robustness. Ensemble achieved F1-score of 0.7652 on diverse datasets.

Advanced ML Frameworks for Hate Speech Detection

0.765+
Best F1-Score

150-400ms
Inference Speed

50+
Languages Supported

94%
User Satisfaction

Transformer Models

> Fine-tune pre-trained transformers for multilingual contexts and low-resource languages.

- IndicBERT V2: For South Asian languages (Hindi, Bengali, Tamil)
- XLM-RoBERTa: Robust multilingual baseline across 100+ languages
- Gemma-2 (9B/27B): Open-source LLMs with few-shot capabilities

Advanced Techniques

> Apply sophisticated training methods to handle class imbalance and edge cases.

- Focal Loss: Focus on hard examples in imbalanced datasets (alpha=0.35, gamma=4.0)
- LoRA Fine-tuning: Parameter-efficient adaptation with rank=16
- ORPO: Preference optimization for instruction-following LLMs

Multi-Modal Integration

> Detect context-dependent hate speech across text, image, and video.

- CNN-RNN: Visual hate symbols + temporal context detection
- Audio Analysis: Detect hate speech in voice content and streams
- Context Fusion: Cross-modal attention mechanisms for holistic understanding

Phase 4: PROTOTYPE - Building Your AI Solution

Rapid MVP Development

Use open-source stacks and pre-trained models to build working prototypes in 24-48 hours. Focus on validation, not perfection.

Tech Stack Essentials

PyTorch + Hugging Face Transformers + FastAPI for scalable, production-ready endpoints with containerization.

Data Strategy

Combine curated datasets (HASOC, Gab, Twitter) with domain-specific annotations. Balance across languages and hate categories.

Model Selection

Start with Gemma-2 or DistilBERT; ensemble 2-3 models for robustness; add few-shot capabilities for rapid adaptation.

Advanced Tools & Frameworks Recommendation

Tier 1: Core Development

- Hugging Face Transformers: Pre-trained models, easy fine-tuning, model hub
- PyTorch Lightning: Scalable training, distributed computing, experiment management
- Weights & Biases: ML experiment tracking, hyperparameter optimization, team collaboration

Tier 2: Deployment & Scaling

- FastAPI + Docker: High-performance API endpoints, containerization, easy deployment
- Ray for Distributed ML: Parallel inference, hyperparameter tuning, real-time model serving
- TorchServe: Production-grade model serving, batching, multi-GPU support

Tier 3: Monitoring & Governance

- Evidently AI: Model monitoring, data drift detection, performance degradation alerts
- Fairness Indicators: Bias auditing across demographics, intersectionality analysis
- SHAP/LIME: Explainability, interpretability for end-users and regulators

Phase 5: TEST - Validation & Iteration Strategy

Quantitative Testing

Evaluate on diverse test sets with stratified sampling. Track Precision, Recall, F1, and AUC across language groups and hate categories.

Confusion matrix analysis reveals false positives (over-censorship) vs. false negatives (missed harm). Target F1 > 0.75 with balanced errors.

Qualitative Testing

Conduct user testing with affected communities, content moderators, and platform teams. Gather feedback on false positives, edge cases.

Perform red-teaming exercises—adversarially test system robustness against bypass techniques, homophone substitution, and code-switching.

Critical Success Factors for Your Hackathon

Design Thinking Excellence

Prioritize empathy and user research over code. Teams winning AI hackathons ground solutions in real user needs, not just technical sophistication.

Bias & Fairness First

- Audit models across demographics (race, gender, language, region).
- Test on code-switched text, regional dialects, and context-dependent expressions.
- Document limitations transparently in your presentation.

User-Centered Explainability

- Provide clear explanations: Why was this content flagged?
- Enable human appeal with context—allow users to provide nuance.
- Build trust through transparency and accountability mechanisms.

Rapid Iteration

Deploy early MVPs, gather feedback, iterate. In 24-48 hours, prioritize validated learning over feature completeness.

Ethical Foundation

Address hate speech thoughtfully. Avoid over-censorship while protecting vulnerable groups. Document your ethical framework clearly.

Clear Metrics

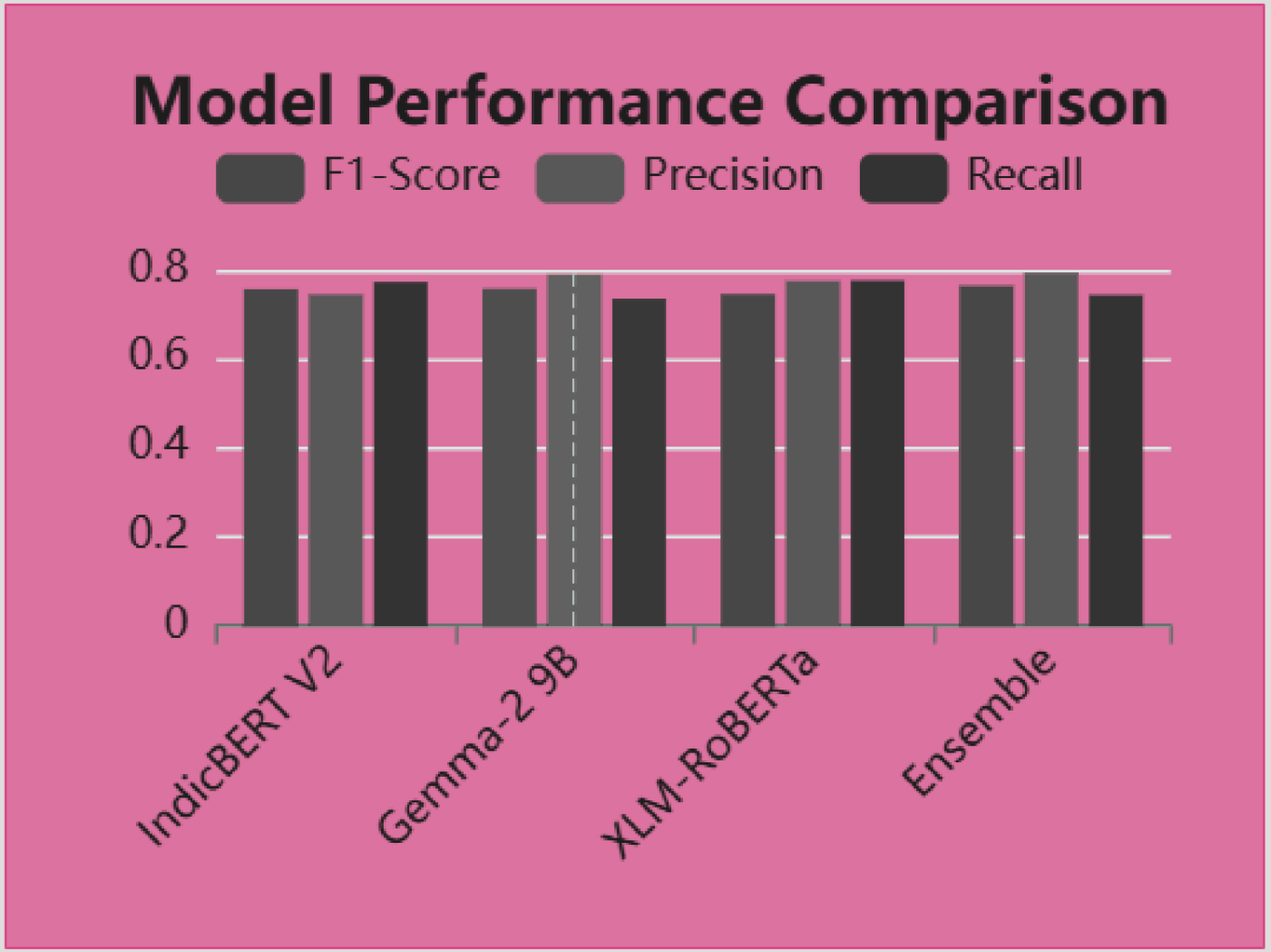
Report F1-score, precision-recall curves, false positive/negative rates. Show how your solution balances accuracy with user safety.

Model Performance Benchmarks & Best Practices

Recommended Baseline

Start with Gemma-2 9B fine-tuned on ensemble with ORPO optimization.
This open-source model achieves F1-score of 0.765+ with strong multilingual support and reduced hallucinations—perfect for hackathon fast-tracking.

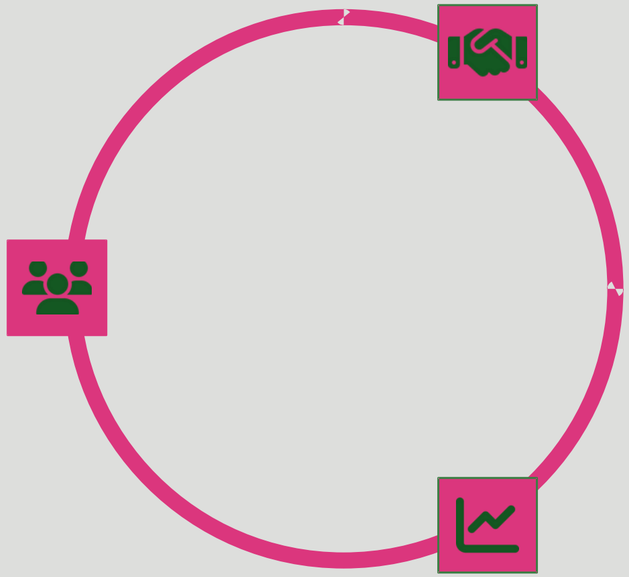
> **Ensemble methods combining IndicBERT V2, XLM-RoBERTa, and Gemma-2 outperform single models. Focal Loss (gamma=4.0) handles class imbalance in sparse hate speech datasets effectively.**



Collaboration & Team Dynamics

Cross-Functional Teams

Combine data scientists, product designers, domain experts in hate speech/social impact, and ethicists. Diverse perspectives strengthen bias detection and fairness in your solution.



Stakeholder Engagement

Involve affected communities early. Partner with civil society organizations, researchers studying hate speech, platform trust/safety teams for real-world validation and credibility.

Agile Sprint Methodology

Daily standups, sprint planning, retrospectives. Allocate: 20% research, 40% model development, 30% testing/iteration, 10% presentation/documentation in your hackathon timeline.

Advanced Implementation Resources

(a)

text	task_1
क्लिक करें और पढ़ें इस स्मार्टफोन के बारे में. Translation - (Click and read about this smartphone.)	NOT
अबे दल्ले कौन बोलता है कि तू पत्रकार है। भड़वा और दलाल है तू।।Translation - (Hey Dalle, who says that you are a journalist. You are a bhadva and a pimp.)	HOF

HASOC-Hindi (2019) dataset sample

(b)

text	task_
सुशिक्षित माणूस सुसंस्कारित असेलच असे नाही. Translation - (An educated person is not necessarily a cultured person.)	NOT
अरे मुर्खा तेच तर सांगताय महापौरांच्या. Translation - (Oh fool, that's what the mayor is saying.)	HOF

HASOC-Marathi (2021) dataset sample

(c)

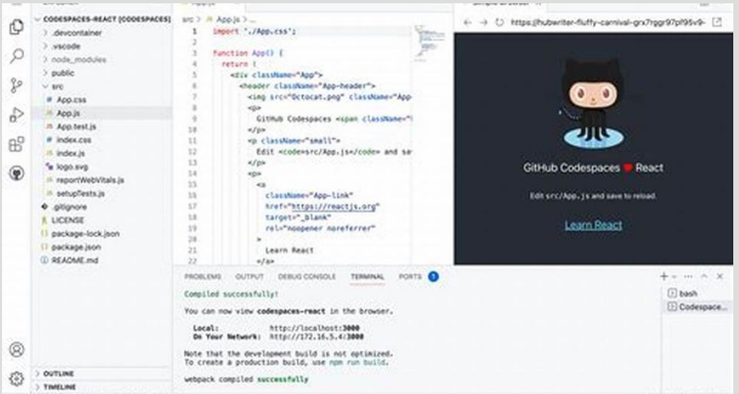
sentence	hate	category
খুব তাড়াতাড়ি রায় কার্যকর করা হোক এইটাই দাবি । Translation - (This is the demand that the judgment should be implemented very soon)	0	crime
তোর কপালে জুতা মারি শালার পুত । Translation - (Son of a rascal, I will thrash you with a shoe on your forehead.)	1	sports

Open-Source Implementations

Leverage transformers.pipeline() for quick baseline, Hugging Face Model Hub for fine-tuned checkpoints, HateCheck benchmark for systematic evaluation across 34 test cases.

Premium Datasets & Benchmarks

HASOC 2023 (Hindi, English, Tamil), Gab Hate Corpus, Twitter HateSpeech, Nisien.ai's Hero platform data. Combine multiple sources for robustness; avoid distribution shift.



Production Deployment Guides

Use Nisien.ai's Hero platform for scaling, deploy with Ray Serve + Kubernetes for horizontal scaling, implement Evidently AI for continuous monitoring and drift detection.

Transform Hate into Hope

Your design-centered AI solution has the power to create safer, more inclusive online communities. Build with empathy. Test rigorously. Impact at scale.

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